

Data Converter

Lecture 5

Calibration Techniques

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电路非理想因素

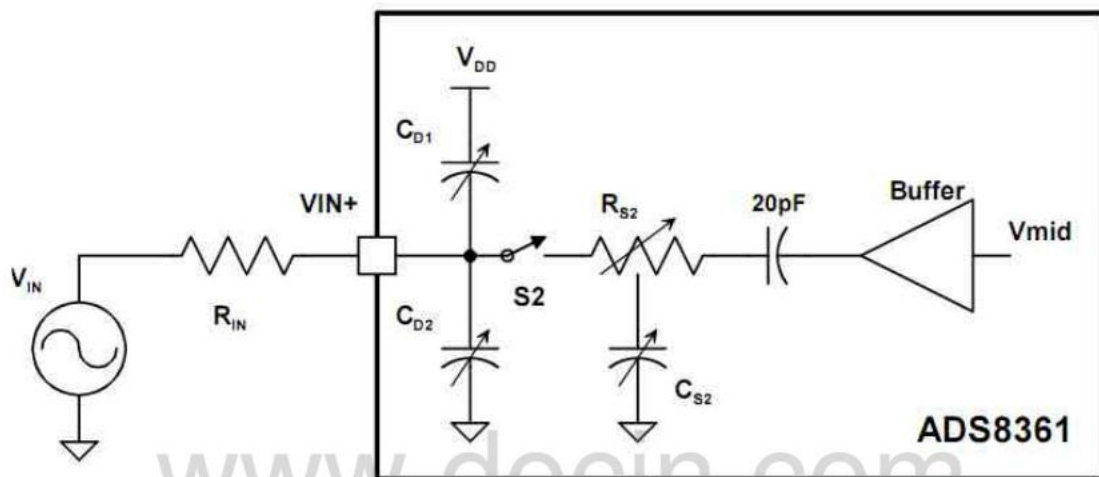
- Capacitor mismatch
- Opamp's finite gain & nonlinearity
- Switch's nonlinear on-resistance
- Incomplete settling
- Offset of opamp & comparator
- Noise & cross-coupled interference
- Clock jitter
- Nonlinear load of the ADC input

非理想效应

- Offset
- Gain error
- Missing code
- Missing level
- Curvature
- Spur & Increased Noise Floor in Spectrum
- Decayed High frequency performance

For Pipeline ADC:
Inter-stage gain error & nonlinearity

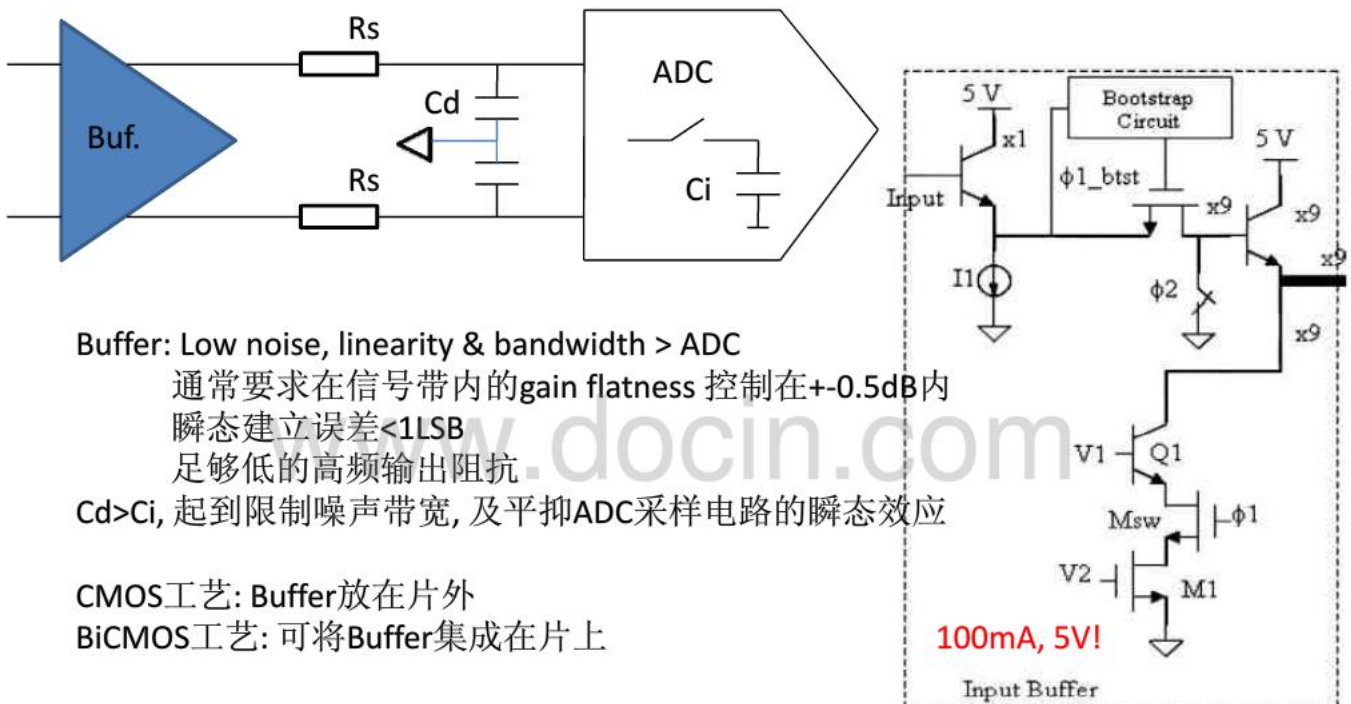
Nonlinearity input load



Equivalent Non-Linear Load of ADC's Input

R_{in} 增大, 将导致更多distortion, 故驱动ADC的buffer要仔细设计
低频下nonlinear load影响不大, 高频下distortion显著增大
差分输入较之单端输入能大幅度降低distortion (>20dB 高频下)

Input Buffer



Buffer: Low noise, linearity & bandwidth > ADC

通常要求在信号带内的gain flatness控制在 $\pm 0.5\text{dB}$ 内

瞬态建立误差 $<1\text{LSB}$

足够低的高频输出阻抗

$C_d > C_i$, 起到限制噪声带宽, 及平抑ADC采样电路的瞬态效应

CMOS工艺: Buffer放在片外

BiCMOS工艺: 可将Buffer集成在片上

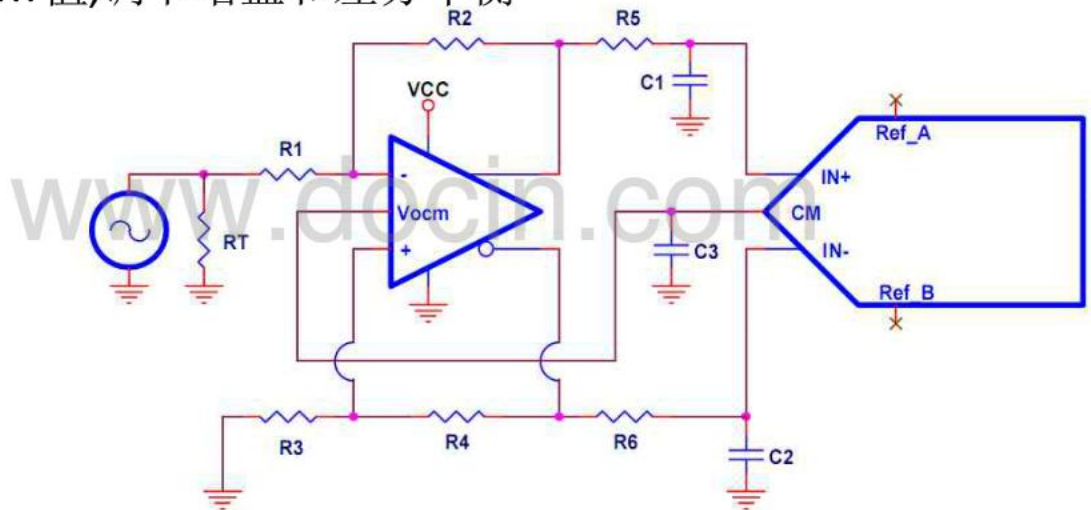
Ref: Ahmed M. A. Ali, etc, "A 14-bit 125 MS/s IF/RF Sampling Pipelined ADC With 100 dB SFDR and 50 fs Jitter", JSSC, 41.8

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ADC DC Coupled input buffer

- Single input, single supply
- Similar to the reference buffer
- R_T ? R_T 与Source resistor并联
- 要调节 R_T 值,调节增益和差分平衡



Calibration techniques

- 按信号域
 - Analog calibration
 - Digital calibration
- 按实时性
 - Foreground calibration
 - Background calibration
- 按校准对象
 - Linear calibration (cap mismatch、finite opa gain)
 - Nonlinear calibration (nonlinearity of opa, switch, incomplete settling, etc.)

Foreground & Background

- **Foreground**
 - 优点：电路简单，校准快
 - 缺点：需要中断正常转换
 - 应用：
 - 可间歇中断转换的应用场合
 - 校准与环境不敏感的非理想因素（如电容失配）
- **Background**
 - 优点：在正常转换的同时进行校准，可实时跟踪环境变化
 - 缺点：电路复杂，或收敛时间长(statistic based scheme)（开始转换后不能马上得到精确结果、生产测试时时间过长影响生产率的问题）
 - 应用：当前高精度转换器的研究热点

Analog calibration for finite gain

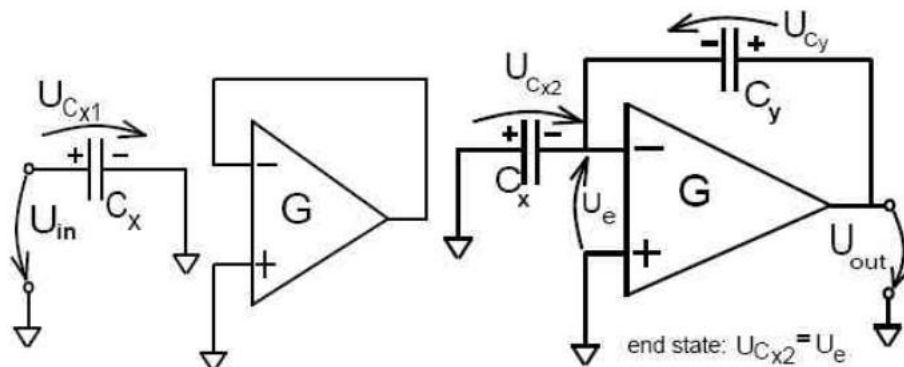
Ref: Reinhard Kindt, "An analog approach to compensate for Opamp offset and finite gain in SC circuitry: A case study of a cyclic RSD ADC"

为何讨论这篇文章:

- 1) 在深亚微米工艺下, 运放的增益和线性成为瓶颈
- 2) 可以校准有限增益和非线性误差, 特别是非线性误差, 即使采用输出校准也很困难
- 3) 本质后台特性

但模拟校准在电路实现上很复杂, 对高速性能不利

Error due to finite gain

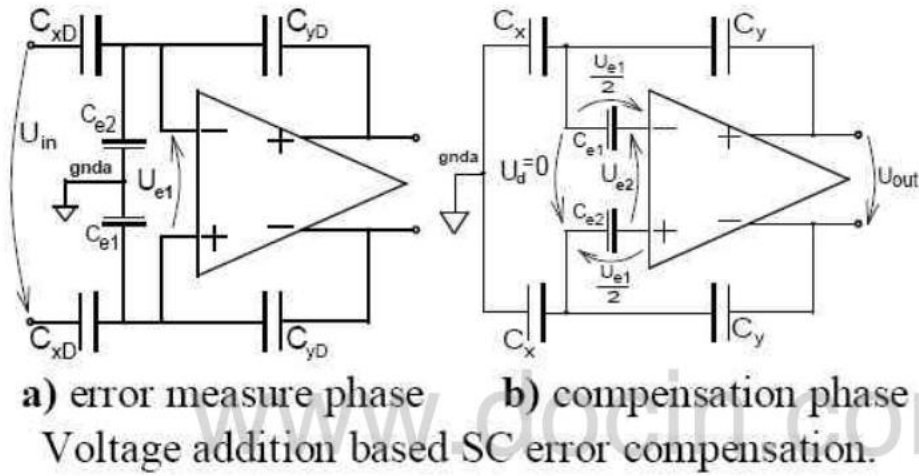


a) sample phase b) charge transfer hold phase

Origin of SC errors due to OpAmp's finite gain.

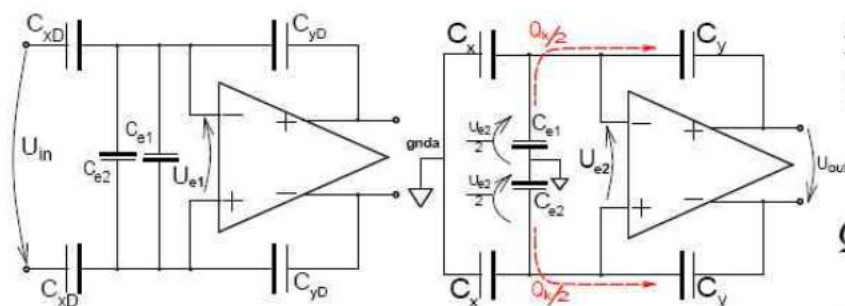
$$\begin{aligned}
 U_{out} &= (U_{in} - U_e) \cdot \frac{C_x}{C_y} - U_e = \\
 &= U_{in} \cdot \frac{C_x}{C_y} - U_e \left(1 + \frac{C_x}{C_y} \right)
 \end{aligned}$$

Voltage addition



Ce1, Ce2采样电压，其大小关系不是很大

Charge addition



这种方法对 C_{e1} , C_{e2} 的值有要求

a) error measure phase b) compensation phase

Charge addition based SC error compensation.

$$Q_k = 2 \cdot C_e \cdot \left(U_{e1} - \frac{1}{2} U_{e2} \right)$$

$$U_k = 2 \cdot \frac{C_e}{C_y} \cdot \left(U_{e1} - \frac{1}{2} U_{e2} \right)$$

$$2 \cdot \frac{C_e}{C_y} \cdot \left(U_{e1} - \frac{1}{2} U_{e2} \right) = U_{e2} \cdot \left(1 + \frac{C_x}{C_y} \right)$$

$$C_e = (C_x + C_y) \cdot \frac{U_{e2}}{(2 \cdot U_{e1} - U_{e2})}$$

$$U_{e1} \approx U_{e2} \approx U_e$$

$$C_e \approx C_y + C_x$$

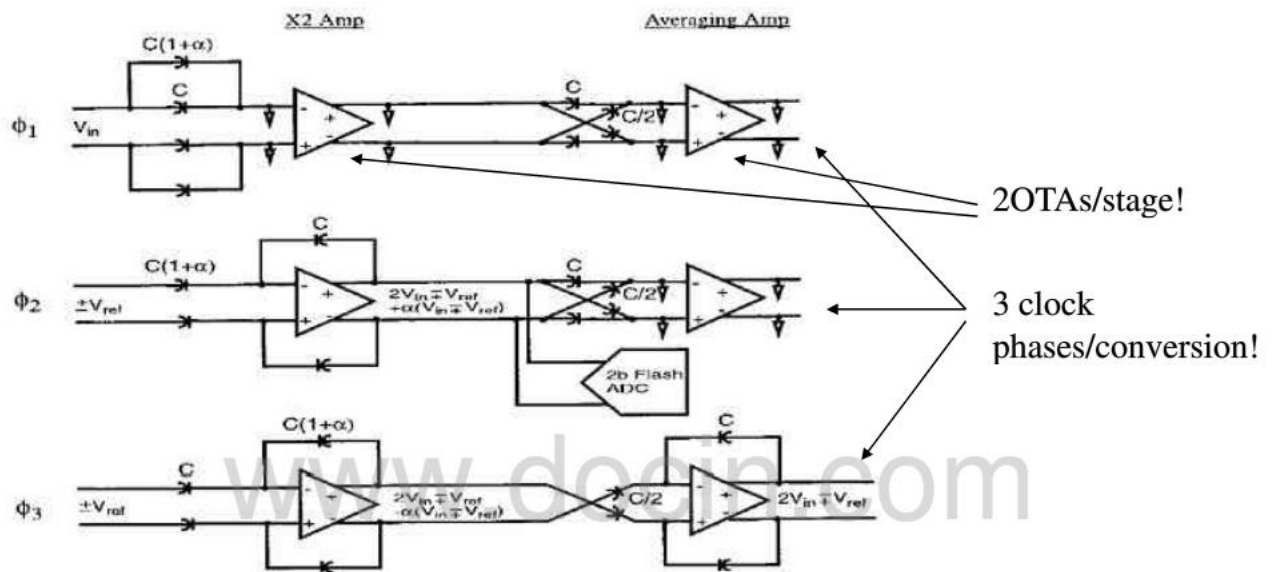
Analog calibration for Capacitor Mismatch

用模拟方法校准电容失配误差曾经有很多人做，
但是，电容失配可用简单的前台数字校准就可
搞定，难的是运放。。。运放，运放，运放！

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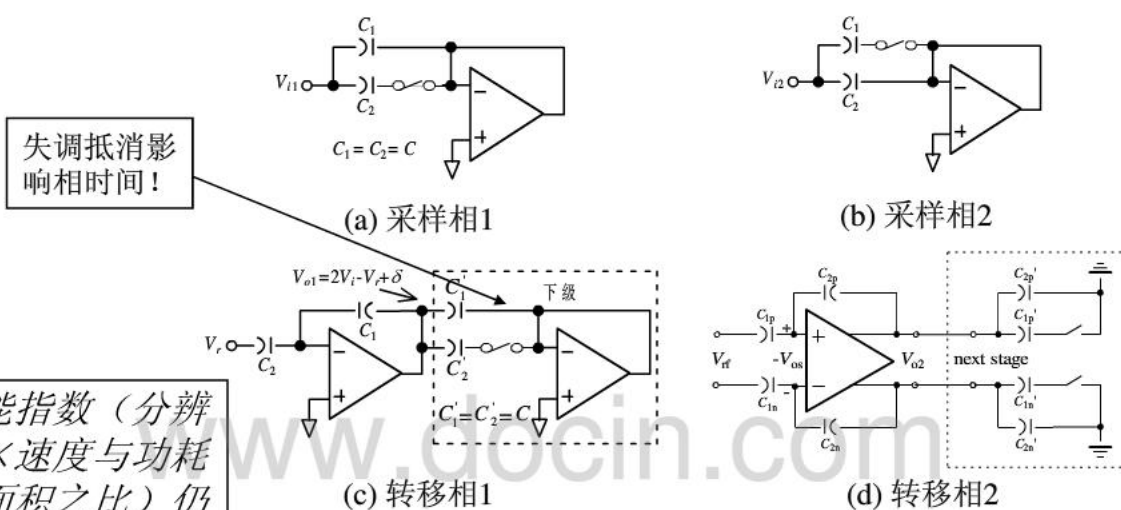
有源电容误差平均方法



Reference:

H S Chen, K Bacrania, and B S Song. A 14b 20Msample/s CMOS Pipelined ADC [A]. ISSCC Dig. Tech. Papers [C], San Francisco, 2000: 46-47

无源电容误差平均技术原理



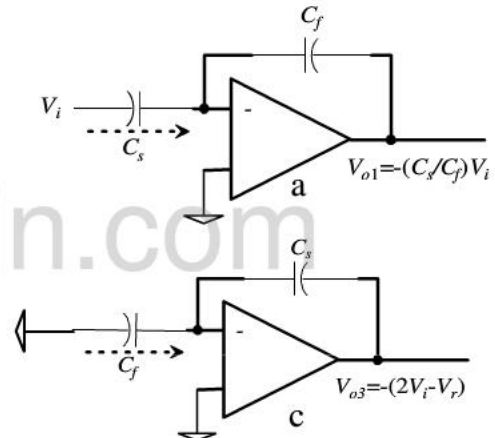
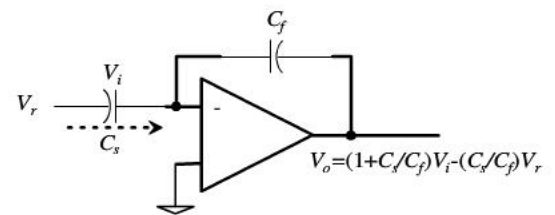
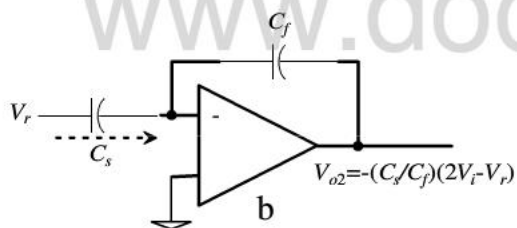
If $V_{i1} = V_{i2} = V_i$, then $V_{o1} = 2V_i - V_{ref} + \delta$ $V_{o2} = 2V_i - V_{ref} - \delta$

电荷共享后 C_1' 和 C_2' 上的等效电压: $V_o = \frac{V_{o1}C_1' + V_{o2}C_2'}{C_1' + C_2'} = 2V_i - V_{ref} + \frac{C_1' - C_2'}{C_1' + C_2'}\delta$

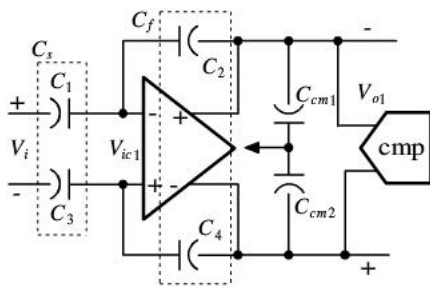
电荷回迁式校准方法的原理

在级电路中，电容失配之所以影响输出精度，是因为采样和反馈在不同的电容上实现，这样等量的电荷在不相等的电容上转移导致了电压的偏差。

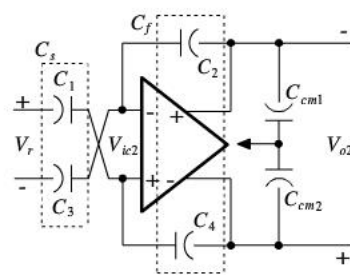
电荷回迁方法的原理：采样和D/A用同一个电容实现，而后采用电容交换技术把电荷转移回采样电容。



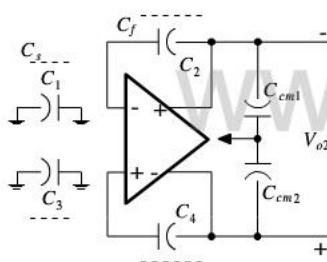
一个转换周期的工作过程



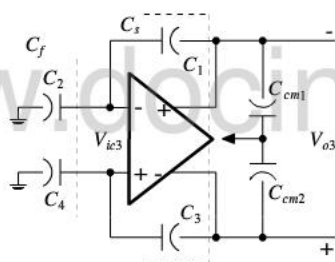
A 采样和A/D转换



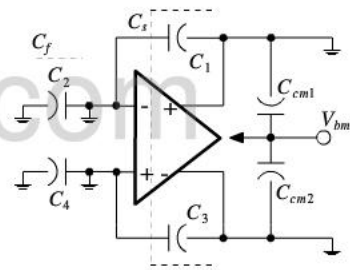
B 放大和D/A转换



C 采样电容复位



D 输出



E 复位

一个转换周期的三相实现

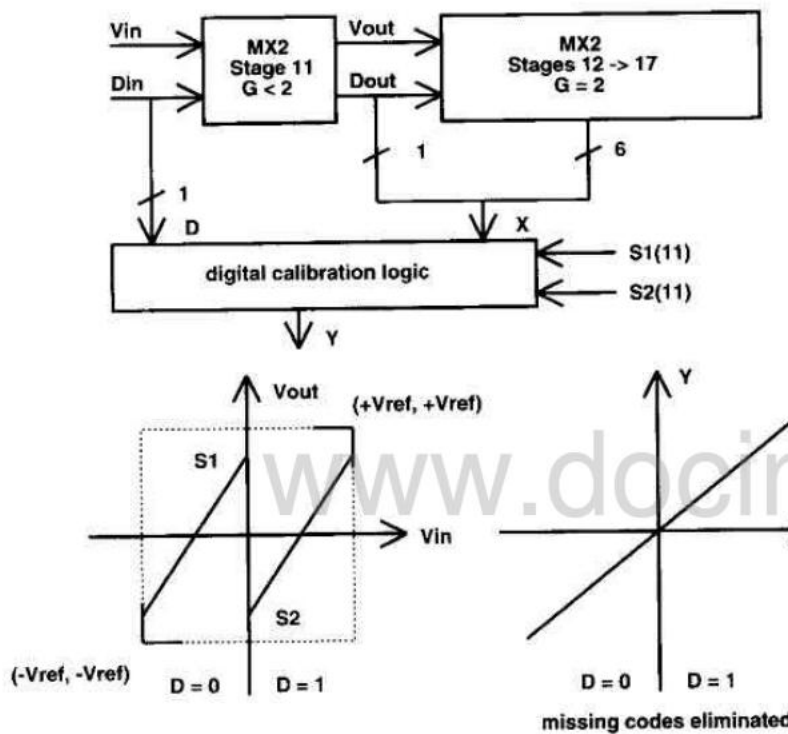
- 采用三相不交叠时钟，C和E在不交叠区内完成
- 不交叠区时间由电容和放电开关导通电阻的RC时常数和建立精度确定
- A/D转换在不交叠区内完成，长不交叠区有利于比较器设计
- 三相时钟对应为A，B，D

各级电路间的时序关系



Stage I-1	Stage I	Stage I+1	Stage I+2
A	B	D	A
B	D	A	B
D	A	B	D

Digital Calibration Principle



$$Y = X, \text{ if } D = 0$$

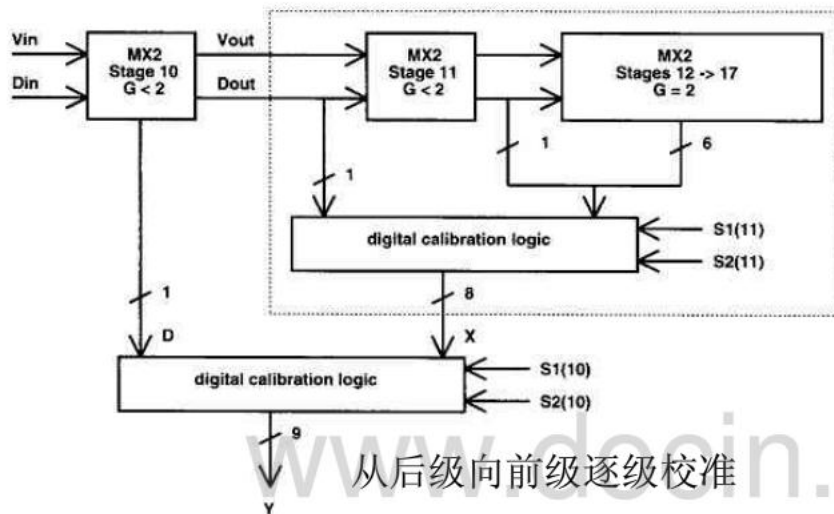
$$Y = X + S_1 - S_2, \text{ if } D = 1$$

➤测量 D 所对应权重，这里须固定 $V_{in}=0$ ，然后令 $D=0$ ，测量对应的输出 $S1$ ，令 $D=1$ ，测量对应的输出 $S2$

➤显然，校准操作需打断正常转换，故是一种前台校准方法

Andrew N. Karanicolas, et al, A
15-b 1-Msample/s Digitally
Self-Calibrated Pipeline ADC.
JSSC 1993

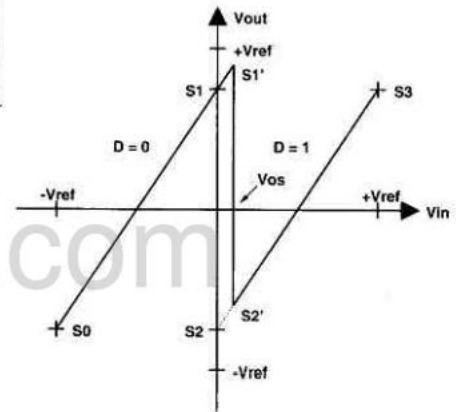
Calibration for higher level stages



是一种前台校准方法，可校准电容失配误差和运放有限增益误差，但运放有限增益误差的校准效果随环境变化而不确定

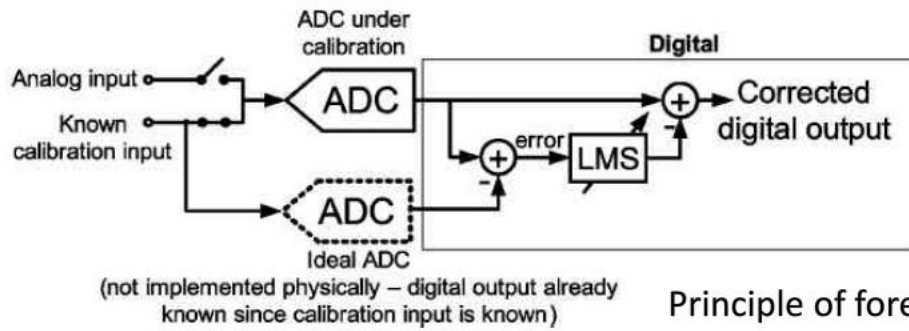
一种高精度ADC设计思路：将运放设计得比较好，采用前台校准方法校准电容失配，电容失配对环境不敏感

为减小测量误差和噪声的影响，S1-S2是多次测量(2048 for example)并平均的结果

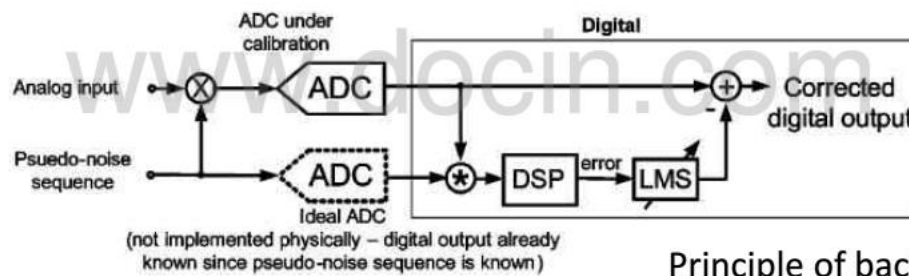


Radix < 2 for comparator offset

Digital Calibration Principle

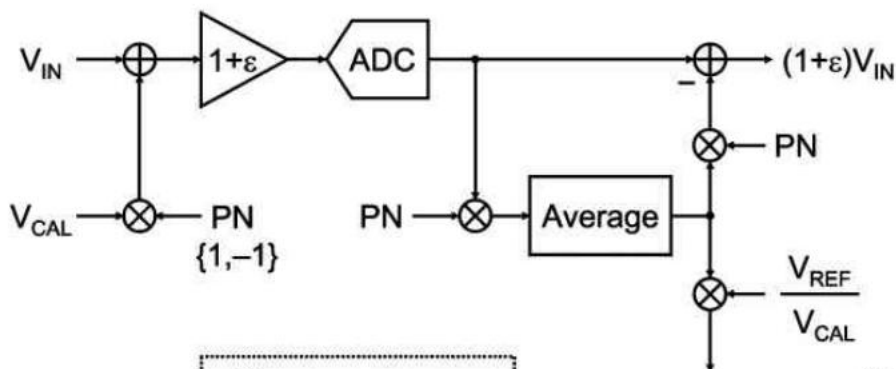


Principle of foreground calibration



Principle of background calibration

Background Calibration



Tradeoff:

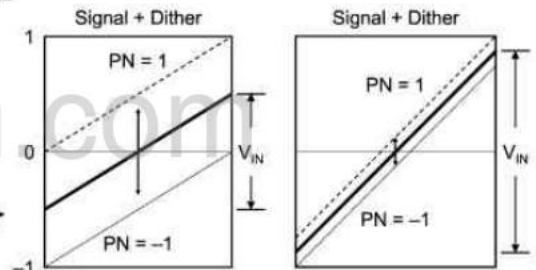
收敛速度: V_{CAL} 尽量大

输出swing: V_{CAL} 尽量小

$$\frac{V_{IN}}{V_{CAL}} PN (1+\epsilon) V_{REF} + PN^2 (1+\epsilon) V_{REF}$$

Average

Noise → Measurement Error



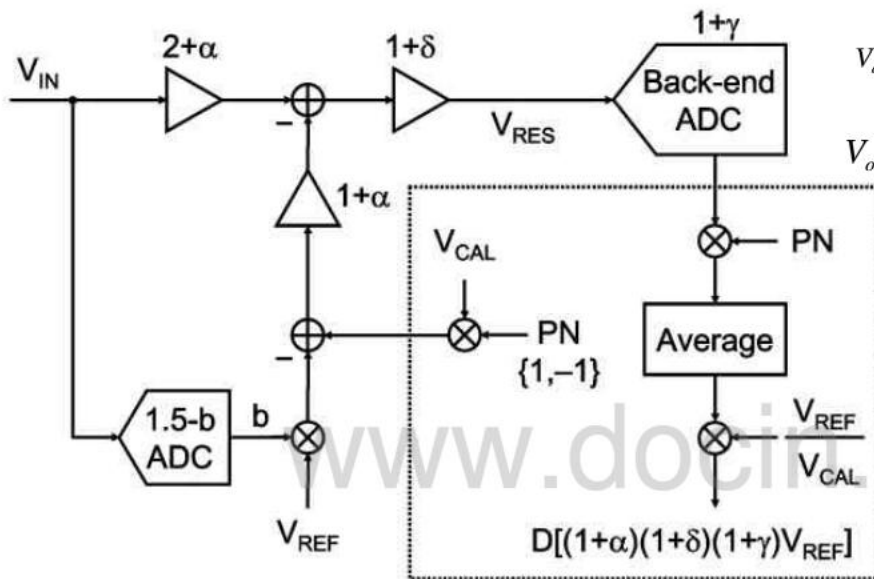
Principle of background gain measurement with PN dithering.

Ref: Yun-Shiang Shu, et al. "A 15-bit Linear 20-MS/s Pipelined ADC Digitally Calibrated With Signal- Dependent Dithering", IEEE JSSC, 2008

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1.5b Stage Calibration



$$V_{out} = \left(\frac{C_f + C_s}{C_f} \cdot V_i - \frac{C_f}{C_f} \cdot V_{ref} \right) \cdot \left(1 - \frac{1}{LG} \right)$$

$$V_{out} = ((2 + \alpha) \cdot V_i - (1 + \alpha) \cdot V_{ref}) \cdot (1 + \delta)$$

$$\alpha = \frac{C_s - C_f}{C_f}$$

$$\delta = -\frac{1}{LG}$$

获取本级权重信息

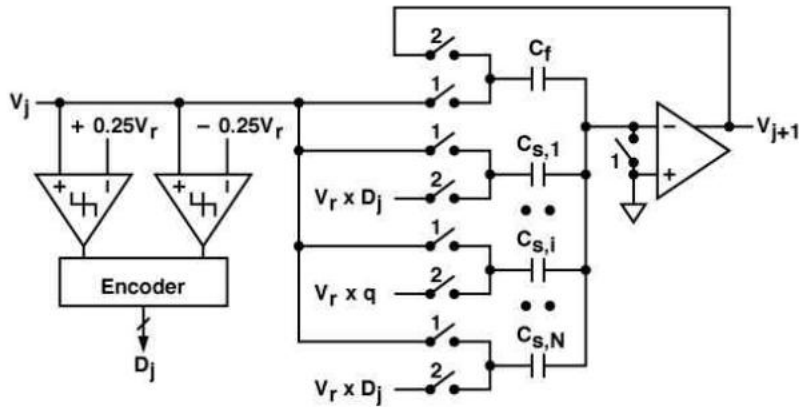
Background gain measurement of an MDAC stage.

Ref: Yun-Shiang Shu, et al. "A 15-bit Linear 20-MS/s Pipelined ADC Digitally Calibrated With Signal- Dependent Dithering", IEEE JSSC, 2008

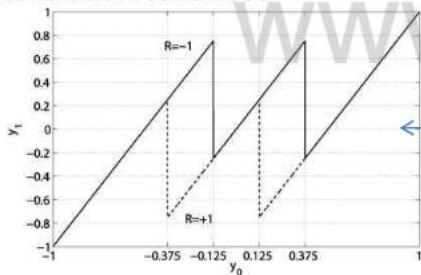
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Vcal的加入



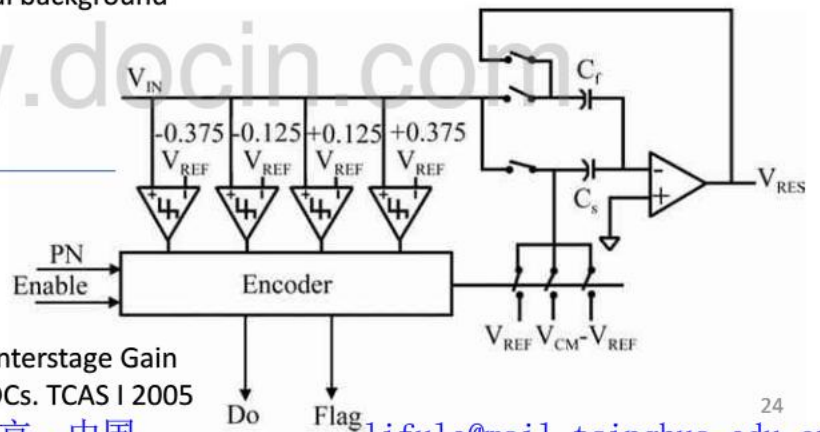
Ref: Liu H C, et al, "A 15-b 40-MS/s CMOS pipelined analog-to-digital converter with digital background calibration. JSSC 2005



Ref: John P. Keane, et al, Background Interstage Gain Calibration Technique for Pipelined ADCs. TCAS I 2005

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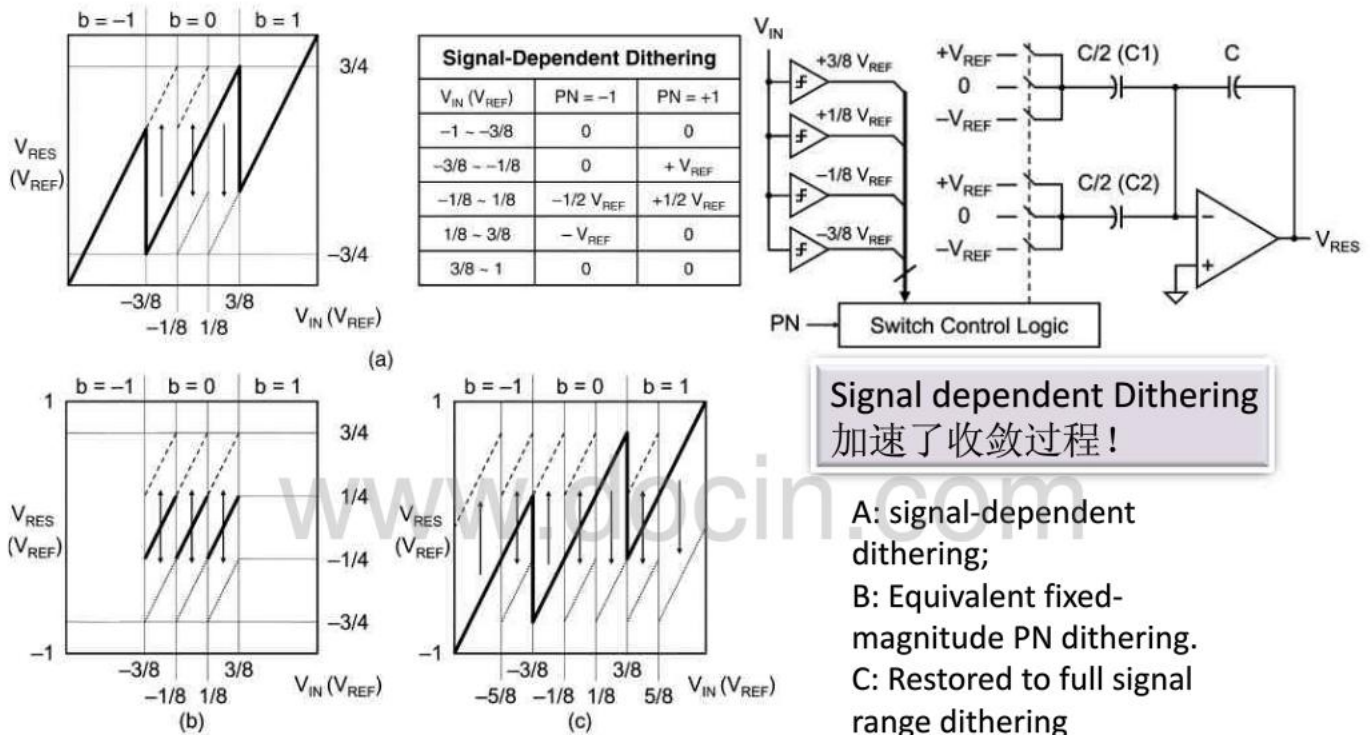
- 1) 可将 C_s 分成 N 块, 在其中一个底板注入 $V_r \times PN$
 N 在校准时间和信号动态范围间进行trade off, 一般取4
- 2) 可将比较器的阈值进行dither, 幅度为 $0.125V_{ref}$
 缺点: 校准对应的输入范围有限, 收敛速度慢



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Signal dependent Dithering



Ref: Yun-Shiang Shu, et al. "A 15-bit Linear 20-MS/s Pipelined ADC Digitally Calibrated With Signal- Dependent Dithering", IEEE JSSC, 2008

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非线性校准

级电路输出电压:

$$r = (1 + \delta g) \left(2^m x - \sum_{i=1}^{2^m-1} D_i V_{ref} \right)$$

$$\delta g = \delta g_0 + \delta g_2 r^2 + \delta g_4 r^4 + \dots + \delta g_n r^n$$

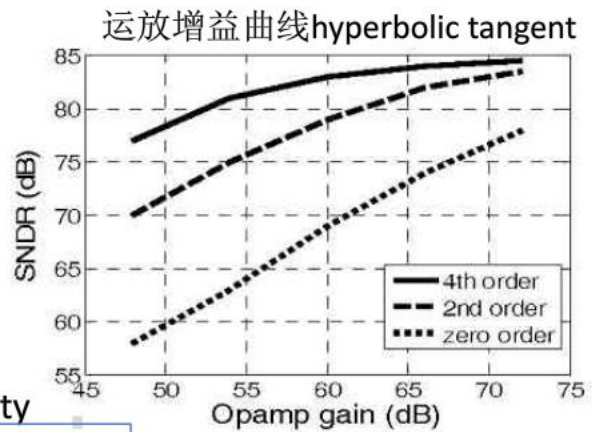
Digital calibration:

Zero order : inter-stage gain error

2nd & 4th order: inter-stage nonlinearity

即使校准了运放有限增益误差和电容失配误差，运放非线性仍然要求运放具有比较高的增益，否则，就要采用非线性校准，但这在计算量和收敛时间上还很不理想

Ref: Mohammad Taherzadeh-Sani, et al, Digital Background Calibration of Interstage-Gain and Capacitor-Mismatch Errors in Pipelined ADCs. 2006



计算量:

Zero order: $cor[PN, Y_{PN}]$

2nd order: $cov[PN, Z, Z^2]$

4th order: $cov[PN, Z, Z^4]$

Lecture 5

END

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